

UofT ML Bootcamp Evaluation 1 - Summer 2026

Hi, this is the first bootcamp evaluation form. Please indicate your answer to the following questions!

* Indicates required question

1. Email *

2. Full Name *

Questions

3. Let's say the 'age' feature is standardized with StandardScaler using $z = (x - \mu) / \sigma$ *. Suppose 'age' has mean $\mu = 30$ and standard deviation $\sigma = 8$. What is the standardized value of a passenger whose age is 46?

Mark only one oval.

☐ 16

☐ -2

☐ 2

☐ 0.5

4. The nominal variable 'color' can be encoded as blue=0, green=1, red=2, or can be one-hot encoded into three 0/1 columns (e.g., [0,0,1]). For a model that treats its inputs numerically, why is one-hot encoding usually preferred over label encoding for a variable like 'color'? *

Mark only one oval.

- ☐ One-hot encoding uses fewer columns than label encoding, so it trains faster and uses less memory.
- ☐ The integers 0, 1, 2 imply an ordering and spacing among the colors that is not real, so a numeric model may treat red as greater than blue.
- ☐ Label encoding can only encode two categories at a time, so three colors cannot be represented.
- ☐ One-hot encoding replaces each color with how frequently it appears, which carries more information.

5. Which of the following plots is good at showing the outliers? *

Mark only one oval.

- ☐ Box Plot
- ☐ Pie Chart
- ☐ Line Chart
- ☐ Heatmaps

6. **What of the following is not an objective of data cleaning? ***

Mark only one oval.

- ☐ Creating new data records.
- ☐ Removing duplicate records.
- ☐ Handling missing data.
- ☐ Correcting inaccurate values

7. **Why do we use feature selection in machine learning? ***

Mark only one oval.

- ☐ To change data into a machine learning format.
- ☐ To pick the best machine learning method.
- ☐ To choose the important features.
- ☐ To fill in missing data.

8. **You've trained a deep decision tree on your dataset and noticed that it performs really well on the training data but poorly on the validation data. ***

Which ensemble technique might help remedy this, bagging or boosting?

Mark only one oval.

- ☐ Bagging
- ☐ Boosting

9. **What is regularization for? ***

Mark only one oval.

- ☐ To speed up model training by reducing the number of computations.
- ☐ To discourage overly complex models so they generalize better to unseen data.
- ☐ To fill in missing values before the model is trained.
- ☐ To increase the model's accuracy on the training set.

10. **Which of the following is NOT intended to regularize an ML model? ***

Mark only one oval.

- ☐ Dropout in a neural network.
- ☐ Adding an L2 penalty on the weights.
- ☐ Standardizing features to zero mean and unit variance.
- ☐ Pruning a decision tree (or capping max_depth)

11. **A new point is at (5, 5). Using Euclidean distance and $k = 3$, which class does k-NN assign it to? ***

Label A data points: (4, 5), (2, 2), (1, 5), (3, 8)

Label B data points: (6, 6), (6, 4), (7, 5), (8, 8)

Mark only one oval.

- ☐ A
- ☐ B

12. **Among those what is not hyperparameters? ***

Mark only one oval.

- ☐ The number of neurons in one layer of neural network
- ☐ Learning rate in gradient descent
- ☐ The coefficient of regularization term
- ☐ The threshold value chosen at each node of a decision tree

13. **A bank subscription classifier is evaluated on 100 clients. Of the 50 clients who actually subscribed (the positive class), the model correctly flagged 40 and missed 10. Of the 50 who did not subscribe, it wrongly flagged 20 and correctly cleared 30. What is the model's RECALL on the positive class?**

Mark only one oval.

- ☐ 0.67
- ☐ 0.7
- ☐ 0.6
- ☐ 0.80

14. **Which technique is commonly used for finding the best hyperparameter tuning ***

Mark only one oval.

- ☐ Feature Enginnering
- ☐ Cross-Validation
- ☐ Data Preprocessing
- ☐ Model Inference

15. **Why cross-validation is more reliable than a single split? ***

Mark only one oval.

- ☐ The model trains on all of the data every time, so no points are ever held out and no luck enters the estimate.
- ☐ Each example is used for validation exactly once and the five fold scores are averaged, so the estimate depends far less on one lucky or unlucky split.
- ☐ It guarantees the tuned model cannot overfit, because averaging over five folds cancels out the variance of the model's learned parameters.
- ☐ Each of the five fits is validated on the four training folds combined, giving every fit a larger and therefore lower-variance test set.

16. **You want the final model to use only a handful of those features (automatic feature selection). Which regularization approach do you want to use? ***

Mark only one oval.

- ☐ Ridge (L2), because squaring the coefficients forces the unimportant ones exactly to zero.
- ☐ Lasso (L1), because it shrinks every coefficient by the same small fraction, keeping all 21 features but making them tiny.
- ☐ Lasso (L1), because its absolute-value penalty pushes weak coefficients exactly to zero, dropping those features from the model.
- ☐ Ridge (L2), because it keeps only the coefficients with the largest absolute values and zeros out the rest.

17. **What problem is k-means++ designed to fix over ordinary k-means?** *

Mark only one oval.

- ☐ k-means++ mathematically guarantees the lowest possible WCSS on any dataset, no matter how the individual points happen to be arranged.
- ☐ Random starts leave the features on different scales, so k-means++ first standardizes every feature to zero mean and unit variance.
- ☐ k-means++ removes the need to fix k beforehand by adding centroids until the within-cluster sum of squares reaches zero.
- ☐ Purely random starts can place two centroids almost on top of each other, which slows convergence and risks a poor final clustering.

18. **Which of the following is not typically considered as an evaluation metrics for classification models?** *

Mark only one oval.

- ☐ Precision
- ☐ Recall
- ☐ Feature Importance
- ☐ F1-Score

19. **What are the two common types of supervised learning? (choose 2)** *

Check all that apply.

- ☐ Classification
- ☐ Clustering
- ☐ Regression

20. Which of the following statements is true about supervised learning? *

Mark only one oval.

- ☐ It is a type of machine learning where the model is trained on labeled data.
- ☐ It is a type of machine learning where the model does not require any training data.
- ☐ It is a type of machine learning where a human must intervene during the training process.
- ☐ It is a type of machine learning that does not require labeled data.

21. For linear regression, which of the following are the inputs (also known as features) that are fed into the model and which the model makes a prediction on? *

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

$Y = X\beta + \varepsilon$

Mark only one oval.

- ☐ x
- ☐ β
- ☐ y
- ☐ ε

22. **What is a gradient? ***

Mark only one oval.

- ☐ The rate of change of the model's accuracy during training.
- ☐ A vector that points in the direction of maximum increase of a function (usually the loss function)
- ☐ A type of data structure used to store multiple values of the same data type for model training
- ☐ A measure of the angle between two intersecting vectors (usually the training dataset and the testing dataset)

23. **What is the purpose of gradient descent? ***

Mark only one oval.

- ☐ Visualizing data patterns
- ☐ Generating random numbers to initialize a model
- ☐ Minimizing the loss function to improve the model
- ☐ Converting categorical data to numerical data

24. **Why do we use ReLU over sigmoid as an activation function? ***

Mark only one oval.

- ☐ Because ReLU outputs a value between 0 and 1, which is easier to interpret.
- ☐ Because sigmoid can make gradient shrinks toward zero, slowing learning in deep networks.
- ☐ Because ReLU is differentiable everywhere, while sigmoid is not.
- ☐ Because ReLU guarantees the network will not overfit.

25. **When training the logistic regression model, we use gradient descent to minimize the loss function. What happens if the loss function became zero?** *

Mark only one oval.

- ☐ This means the model's prediction will always be zero.
- ☐ This means our code is wrong, because it is impossible.
- ☐ This means the model now fit the training dataset really well.
- ☐ This means the model now fit the test dataset really well.

26. **The output of logistic regression is:** *

Mark only one oval.

- ☐ A continuous value
- ☐ A probability score
- ☐ A categorical label
- ☐ A binary value

27. **How is the splitting attribute chosen in a decision tree?** *

Mark only one oval.

- ☐ Randomly selecting an attribute from the dataset.
- ☐ Choosing the attribute that has the highest information gain or the lowest Gini impurity.
- ☐ Using gradient descent to find the best attribute.
- ☐ Selecting the attribute that has the highest correlation with the target variable.

28. **What is the purpose of a cost function? ***

Mark only one oval.

- ☐ The cost function is used to calculate the total number of data points in the training set.
- ☐ The cost function is a measure of how good or bad the decision tree fits the training data.
- ☐ The cost function is used to evaluate the complexity of the decision tree.
- ☐ The cost function is only applicable to unsupervised learning algorithms, not decision trees.

29. **What is ensemble learning? ***

Mark only one oval.

- ☐ A single machine learning model
- ☐ A group of unrelated models
- ☐ A technique that combines predictions from multiple models
- ☐ A method to visualize data

30. **In Random Forest, what is bootstrapping used for? ***

Mark only one oval.

- ☐ Enhancing model interpretability
- ☐ Preventing overfitting and speed up training
- ☐ Generating different random subsets of data for training
- ☐ Improving the convergence of the algorithm

31. **Why are activation functions important in MLPs? ***

Mark only one oval.

- ☐ They allow for the introduction of linear transformations.
- ☐ They determine the number of layers in the network.
- ☐ They introduce non-linearity, enabling the model to learn complex patterns.
- ☐ They reduce the computational complexity of the network.

32. **Which of the following is NOT a property of the sigmoid function? ***

Mark only one oval.

- ☐ Differentiable for all real numbers
- ☐ Output range between 0 and 1
- ☐ The result of $\text{sigmoid}(2000)$ is twice as large as $\text{sigmoid}(1000)$
- ☐ Continuous for all real numbers

33. A single sigmoid neuron takes input $x = 1$, with weight $w = 1$ and bias $b = -1$, so that $z = wx + b$. The output is $a = \sigma(z)$, and the cost is $C = \frac{1}{2}(a - y)^2$ with target $y = 1$. *

You are given the element derivatives:

$$\partial C / \partial a = a - y$$

$$\partial a / \partial z = \sigma'(z) = \sigma(z)(1 - \sigma(z))$$

$$\partial z / \partial w = x$$

Using the chain rule, what is $\partial C / \partial w$?

Mark only one oval.

- ☐ -0.5
- ☐ -0.125
- ☐ +0.125
- ☐ -0.0625

34. Which of the following best describes a Multi-layer Perceptron (MLP)? *

Mark only one oval.

- ☐ A single-layer feedforward neural network.
- ☐ A neural network that primarily uses convolutional layers.
- ☐ A type of recurrent neural network optimized for time-series data.
- ☐ A feedforward neural network with one or more hidden layers.

35. **Which of the following is a primary reason for using convolutional layers in a neural network when processing image data?** *

Mark only one oval.

- ☐ They increase the number of parameters in the network.
- ☐ They allow the network to be fully connected.
- ☐ They are designed to detect local patterns, like edges and textures.
- ☐ They ensure that the network only processes black and white images.

36. **In a CNN, what does the term 'pooling' generally refer to?** *

Mark only one oval.

- ☐ Increasing the depth of the network by adding more convolutional layers.
- ☐ The process of periodically saving the network's weights to avoid data loss.
- ☐ Reducing the spatial dimensions of the data by taking the average or maximum value from a group of neighboring pixels.
- ☐ Applying a filter to increase the sharpness of the image before processing.

37. **Which of the following is NOT a common activation function used in CNNs?** *

Mark only one oval.

- ☐ ReLU (Rectified Linear Unit)
- ☐ Tanh
- ☐ Leaky ReLU
- ☐ Softmax

38. **In the context of CNNs, what does a 'stride' refer to? ***

Mark only one oval.

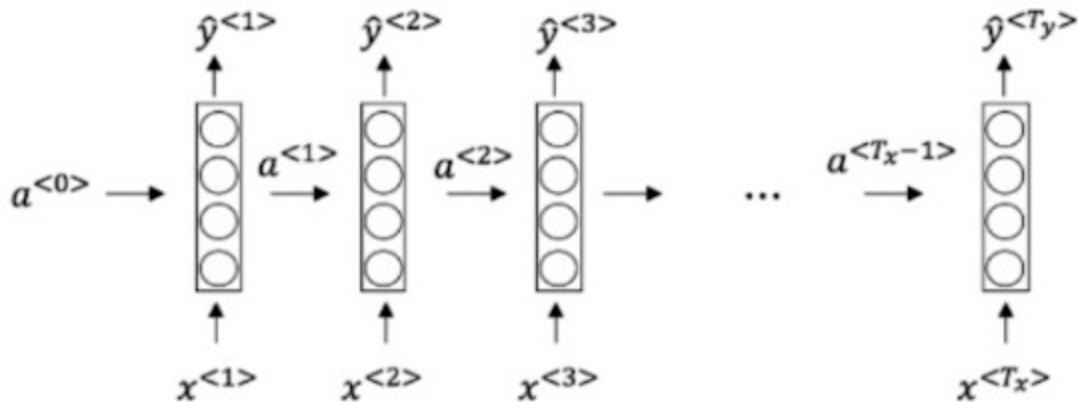
- ☐ The depth of the convolutional filter.
- ☐ The size of the pooling window.
- ☐ The number of layers in the network.
- ☐ The number of pixels the convolutional filter moves after processing a section of the input.

39. **In a Recurrent Neural Network, what is the primary purpose of the hidden layer? ***

Mark only one oval.

- ☐ To process the input data and produce the final output.
- ☐ To store sequential information for memory
- ☐ To apply activation functions to the input features.
- ☐ To directly connect with the output layer.

40. **Suppose you have the following RNN architecture. In what cases would you use it?** *



Mark only one oval.

- ☐ Action Recognition in a video
 - ☐ Sentiment Classification
 - ☐ Image Recognition
 - ☐ Named Entity Recognition
41. **What is the main advantage of using a Recurrent Neural Network (RNN) compared to a traditional feedforward neural network?** *

Mark only one oval.

- ☐ RNNs can predict on input sequences of any length
- ☐ RNNs have more layers, leading to better performance.
- ☐ RNNs are faster in training due to parallelization.
- ☐ RNNs require less memory for computation.

42. **To which of the following you would use many-to-one architecture of RNN? ***

Mark only one oval.

- ☐ Sentiment Classification
- ☐ Question Answering
- ☐ Named Entity Recognition
- ☐ Music Generation

43. **You are training an RNN based architecture and realized during the training that the weights have a value of NaN. Which of these is a possible reason for the same? ***

Mark only one oval.

- ☐ Because the RNN cell uses sigmoid activation function.
- ☐ Because it uses ReLU activation function.
- ☐ Vanishing gradient problem
- ☐ Exploding gradient problem

44. Which of the following loc commands correctly selects the row with the name 'Charlie' and all columns? *

	Name	Age	Score
0	Alice	25	90
1	Bob	18	82
2	Charlie	12	70
3	David	30	95

Mark only one oval.

- ☐ df.loc[2]
- ☐ df.loc['Charlie']
- ☐ df.loc[df['Name'] == 'Charlie']
- ☐ df.loc[2, :]

45. **Fill in the blank to specify the hyperparameter grid correctly:** *

```
from sklearn.svm import SVC
from sklearn.model_selection import GridSearchCV
```

```
# Initialize the SVM classifier
svm_classifier = SVC()
```

```
# Define the hyperparameter grid
param_grid = {
    'kernel': ['linear', 'rbf', 'poly', 'sigmoid'],
    'C': [0.1, 1, 10, 100],
    'gamma': [1, 0.1, 0.01, 0.001]
}
```

```
# Create the GridSearchCV object
grid_search = GridSearchCV(svm_classifier, _____, cv=5)
```

Mark only one oval.

- ☐ param_grid
- ☐ grid
- ☐ hyperparameters
- ☐ parameters

46. **When preparing a dataset for logistic regression to classify items into two groups (e.g., real vs. fake), which of the following steps is essential?** *

Mark only one oval.

- ☐ Converting numeric features into categorical features.
- ☐ Normalizing all text data using NLP techniques.
- ☐ Splitting the dataset into training and testing subsets.
- ☐ Aggregating all features into a single composite score.

47. **In a binary classification problem like distinguishing between two classes (e.g., real vs. fake), what is typically the role of the target variable?** *

Mark only one oval.

- ☐ To provide a continuous score representing the condition of each item.
- ☐ To indicate the specific category or class to which each item belongs.
- ☐ To list all possible categories for items in the dataset.
- ☐ To summarize the features of the items into a single descriptor.

48. **What is the main reason for using a loop to make predictions on a test set in machine learning?**

*

Mark only one oval.

- ☐ To enhance the model's performance with each new prediction.
- ☐ To apply the prediction function to each individual test sample and collect the results.
- ☐ To decrease the likelihood of overfitting by evaluating multiple subsets.
- ☐ To recalibrate the model parameters before making each prediction.

49. **An input image is 28×28. A convolutional layer applies a 3×3 kernel with stride 1 and no padding. What is the spatial size of the output feature map?**

*

Mark only one oval.

- ☐ 28 X 28
- ☐ 26 X 26
- ☐ 25 X 25
- ☐ 14 X 14

50. **What does specifying a kernel size of 3×3 and an activation function in a convolutional layer typically achieve?**

*

Mark only one oval.

- ☐ It determines the compactness of the model's representation and the threshold for activation.
- ☐ It sets the filter dimensions for feature detection and applies a nonlinear transformation to the feature maps.
- ☐ It controls the rate of learning and the depth of the network's architecture.
- ☐ It aligns the data inputs with the expected outputs by resizing the feature maps.

51. **In constructing a basic feedforward neural network manually for predicting temperature, what are the essential components that need to be defined for the hidden layer?**

*

Mark only one oval.

- ☐ Hidden weights and biases, used to calculate the output from the input layer before applying an activation function.
- ☐ Hidden activation function only, as weights and biases are automatically adjusted.
- ☐ Output weights and biases, as these are crucial for transferring data from the input to the hidden layer.
- ☐ The learning rate and loss function, which are critical for the initial setup of each layer.

52. **What is the main limitation of the Bag-of-Words representation? ***

Mark only one oval.

- ☐ Requires very deep networks to learn semantics
- ☐ Ignores word order and context
- ☐ Produces dense low-dimensional vectors
- ☐ Cannot be used for classification tasks

53. **Which problem does attention alignment first address in Seq2Seq RNNs? ***

Mark only one oval.

- ☐ Vanishing gradients in recurrent cells
- ☐ Parallelism on GPUs
- ☐ Focusing the decoder on relevant encoder steps
- ☐ Tokenization of rare words

54. **A classic Transformer block follows which order? ***

Mark only one oval.

- ☐ Feed-Forward → Self-Attention → Add&Norm
- ☐ Self-Attention → Add&Norm → Feed-Forward → Add&Norm
- ☐ LayerNorm → Self-Attention → Feed-Forward
- ☐ Add&Norm → Self-Attention → Feed-Forward

55. **Why are positional encodings required in vanilla self-attention?** *

Mark only one oval.

- ☐ To reduce model size
- ☐ To normalize gradient flow
- ☐ Because self-attention is permutation-invariant without them
- ☐ To replace token embeddings for rare words

56. **Which Transformer variant is typically used for unidirectional language generation?** *

Mark only one oval.

- ☐ Encoder-only (BERT)
- ☐ Encoder-decoder (T5)
- ☐ Decoder-only (GPT-style)
- ☐ Recurrent decoder with attention

57. **The standard pre-training objective for GPT-style models is:** *

Mark only one oval.

- ☐ Masked-language modeling (MLM)
- ☐ Machine translation loss
- ☐ Next-token (causal) prediction with cross-entropy
- ☐ Contrastive sentence embedding loss

58. **One advantage of self-attention over RNNs is: ***

Mark only one oval.

- ☐ Lower parameter count in all cases
- ☐ Ability to model infinite context
- ☐ Full sequence parallelism during training
- ☐ No need for positional information

Coding Questions

59. **Complete the blanks below to enable the code to run successfully.**

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.svm import SVC

# Load the Iris dataset
data = load_iris()
X, y = data.data, data.target

# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

# Initialize the SVM classifier
svm_classifier = SVC()

# Define the hyperparameter grid for GridSearchCV
param_grid = {
    'kernel': ['linear', 'rbf', 'poly', 'sigmoid'],
    'C': [0.1, 1, 10, 100],
    'gamma': [1, 0.1, 0.01, 0.001]
}

# Create the GridSearchCV object with cross-validation (e.g., 5-fold cross-
validation)
grid_search = GridSearchCV(_____, _____, cv=_____)

# Perform the hyperparameter tuning on the training data
grid_search.fit(X_train, y_train)

# Get the best hyperparameters from the GridSearchCV
best_params = grid_search.best_params_
print("Best hyperparameters:", best_params)
```

60. **Please complete code by replacing all `### EXERCISE CODE HERE` with your code.**

*

```
import pandas as pd

import numpy as np

# weight-height dataset

dataset =
pd.read_csv("https://gist.githubusercontent.com/nstokoe/7d4717e96c21b8ad04
ec91f361b000cb/raw/bf95a2e30fceb9f2ae990eac8379fc7d844a0196/weight-
height.csv")

gender_mapping = {'Male': 0, 'Female': 1}

dataset["Gender"] = dataset["Gender"].map(gender_mapping)

display(dataset)

#put the columns of ones into the front of the dataset

dataset.insert(0, 'intercept', 1) ### EXERCISE CODE HERE

X = dataset[ ['intercept',"Gender","Height"] ].values

y = dataset["Weight"].values

# Calculating the coefficients using the normal equation:  $(X^T * X)^{-1} * X^T * y$ 

coefficients = np.linalg.inv(X.T @ X) @ X.T @ y ### EXERCISE CODE HERE

print(coefficients)

"""predict the weight of someone with height of 70 and who have gender `male`"""

X_test = np.array([1,0,70])
```

Making predictions for the test set

y_prediction = X_test @ coefficients

print(y_prediction)

61. **Apply logistic regression to the following dataset, to classify bank notes as real or fake.** *

#The Banknote Authentication Dataset is a collection of data used for binary classification. It contains features extracted from images of genuine and counterfeit banknotes, captured using various sensors.

```
import pandas as pd
dataset =
pd.read_csv("https://raw.githubusercontent.com/Dataweekends/zero_to_deep_learning_video/master/data/banknotes.csv")
display(dataset)
```

```
X = dataset.drop(columns=['class']) # Replace 'target_column_name' with the
actual target column name
y = dataset['class']
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
X_train, X_test, y_train, y_test = X_train.to_numpy(), X_test.to_numpy(),
y_train.to_numpy(), y_test.to_numpy()
```

EXERCISE CODE HERE

62. **Write a feedforward function that works on neural networks of 5 layers. ***

```
def feedforward(a, biases, weights):
    #layer 1 --> 2
    b = biases[0]
    w = weights[0]
    a_layer2 = sigmoid(np.dot(w, a) + b)

    # EXERCISE code
```

63. **You are building a simple CNN using TensorFlow's Keras API to classify images. ***
Fill in the blank to add a convolutional layer to the model:

```
import tensorflow as tf

model = tf.keras.Sequential()

# Add a convolutional layer with 32 filters, a kernel size of (3,3), and 'relu'
activation:
model.add(tf.keras.layers.____(32, (3, 3), activation='relu'))
```

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